

3 Developing Web-based GIS/MCE: Improving Access to Data and Spatial Decision Support Tools

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Introduction

Much research has focused in recent years on the role of geographic information systems (GIS) in spatial decision-making and support (e.g., Clarke, 1990; Kyem, 1994; Fedra, 1995; Carver et al., 1997). Some research has focused more specifically on developing GIS-based multi-criteria evaluation (MCE) techniques for improving spatial decision-making where complex trade-offs between multiple and conflicting criteria are required (e.g., Janssen and Rietveld, 1990; Carver, 1991a; and Eastman et al., 1993). Without application within a GIS framework, many multi-criteria analysis tools lack sufficient flexibility to handle spatial data and are often inaccessible to potential users. Accordingly, much research effort has been directed at integrating GIS and MCE technologies and developing these into effective decision-support tools. In developing integrated GIS and multi-criteria techniques for controversial site search and evaluation problems such as nuclear waste disposal facilities, Carver (1991a, p.337-8) describes the potential for such systems as follows:

...A PC or workstation-based GIS/MCE system and an experienced operator in a committee room could create significant improvements in the way decisions for siting are made. In addition... SDSS (spatial decision support systems) may also have an important role to play in providing more efficient means of public participation and consultation throughout the site-search process by allowing... feedback to decision-makers regarding public sentiment.

It is interesting to note that these ideas, although not unique, were being developed long before the advent of the World Wide Web (WWW or 'web'). As such, these systems were, by necessity, bound to stand-alone or

locally networked computers, which severely restricted their outreach to interested individuals or stakeholder organizations. Current thinking, however, involves a move away from stand-alone systems towards developing open access spatial decision support systems on the web. In creating on-line systems it is possible to greatly widen the potential audience and, hopefully, increase public involvement in spatial decision-making. At the same time, the open nature of the web may be exploited to develop a level of independence from traditional decision-making groups and bodies.

This chapter explores the development of GIS/MCE systems on the web and evaluates their utility as mechanisms for improving public participation in spatial decision making. The usefulness of the web as a delivery mechanism for GIS/MCE systems is considered before developments in on-line GIS are described. Current GIS/MCE systems on the web are reviewed together with practical problems arising from their application. The potential role of on-line GIS/MCE systems for improving public participation in environmental decisions is discussed with particular reference to what extent they can supplement existing planning procedures.

GIS and the WWW

The rise of the Internet and the World Wide Web has created many opportunities for those involved in GIS research. Over the last few years, many GIS sites have appeared on the web giving the general public access to information, systems, and data. The content of GIS web sites ranges considerably from those providing on-line access to scientific papers to advanced real-time GIS systems. Table 3.1 lists the basic categories of GIS web site and gives examples. These are split into four categories based on the type of information flows and the opportunity for interaction with the GIS and appropriate spatial datasets.

The most basic category of GIS web sites listed in Table 3.1 are those disseminating information about specific GIS projects and their results. These take the form of on-line papers, reports, and GIS output. The second category includes those web sites providing more general information about GIS and spatial data. These include dedicated GIS web sites listing other GIS-related web sites, Frequently Asked Questions (FAQs), GIS glossaries, and spatial data archives. The first two categories are characterized by one-way information flows; from host server (information provider) to client (information user). The third category of GIS web site

includes those sites that provide a level of interaction with GIS software and data. These include interactive software, project demonstrators, and GIS tutorials. The fourth and final category of GIS web site includes the sites whose main aim is to provide some form of GIS-based service, be this spatial data retrieval, decision support and collaborative systems. These third and fourth categories therefore emphasize two-way information flows between server and client and client and server.

Table 3.1 Categorization of GIS web pages

Category	Content	Example
Dissemination	Project description Reports, papers, etc.	Boulder County Land Use GIS NCGIA conference proceedings
Information	FAQs GIS lists Glossaries Data archives	U.S. Census Bureau GIS FAQ Edinburgh University, Geography Department Jefferson County GIS Glossary Global Resources Information Database
Interaction	Demonstrations Tutorials	REGIS San Francisco Bay GIS ESRI Arc/Info on-line tutorial
Provision	Information retrieval Decision support Collaborative systems	East St. Louis Geographic Information Retrieval System Open Spatial Decision-making GeoMed Project

With the increased availability of GIS and GIS databases on the web, previous criticism of GIS as an elitist technology (Pickles, 1995) may no longer be valid. Indeed, we are now beginning to witness the popularization of GIS, particularly within business and government. This has come about partly through the GIS applications seen on the web and partly through the incorporation of selected GIS mapping functionality within standard productivity software such as spreadsheets and within specialist packages such as atlases and navigation aids. However, despite these developments, GIS is still a long way from being a universal technology.

Most GIS-related sites on the web fall within the first two categories in Table 3.1, in that they only disseminate GIS research and/or serve to share GIS-relevant data and information. Of the remaining sites, the majority is demonstrations using sample data. Expertise in the use of GIS software and interpretation of spatial data may be lacking on the part of casual users, however, restricting the full and effective use of the system, while the delivery medium itself may not be wholly appropriate for public use (Carver, 1997). Furthermore, very few web-based systems allow the user to populate the data space with new data and, therefore, do not fall into the fourth category described in Table 3.1.

A number of GIS-based SDSS are starting to appear on the web. A couple of notable examples are the East St. Louis Action Research Project (ESLARP) (1998) and the GeoMed Project (1996) (Geographic Mediation System). ESLARP is a local participatory planning system. This incorporates the GIS-based East St. Louis Geographic Information Retrieval System (EGRETS) which provides a gateway to geographic information for use in planning decisions regarding the U.S. city of East St. Louis, Illinois. A quick tour of the EGRETS system gives a flavor of its capabilities. The 'Explore' tool gives access to different types of geographical information such as census data, land use information, and street maps. The spatial datasets provided are all fully searchable via the 'Atlas' tool. A novel feature is the 'Studio' tool which allows the design and creation of new maps using Java applets. The GeoMed service is a collaborative spatial decision support system aimed at:

Improving ways to distribute heterogeneous geographical information and Geographical Information Systems. The main goal of the project is to make geographical information more accessible and affordable and to help open and democratise public policy and planning procedures. This will improve the quality and acceptability of decisions and reduce the considerable expense of unnecessary delay and legal conflicts. GeoMed will integrate services for creating, distributing and using geographical information in the sectors of urban planning, public administration and environmental protection. (GeoMed Project, 1996)

GeoMed works by providing information in a common format, by providing a shared workspace for geographical information handling using standard document formats (SGML and OpenDoc), and by providing mediation services such as 'round table' discussions to assist in group planning and decision-making. Currently four demonstrator projects are working within the GeoMed project: the City of Bonn in Germany, Tilburg

in the Netherlands, the Tuscany Region in Italy, and the Technical Chamber of Greece.

One example developed by the author and colleagues at the University of Leeds is the Open Spatial Decision-making (OSDM) web-site (Carver et al., 1996; 1997). This is a simple information system and GIS-based SDSS focusing on the controversial problem of where to dispose of radioactive waste in Britain. The system uses spatial datasets and spatial analytical tools to help explore the decision problem and develop informed personal siting decisions. The basic objectives behind the development of the system are to experiment with web design for SDSS, inform the public about a particular spatial decision problem and study the effects of locality, scale and social context on decisions made.

GIS/MCE and the WWW

Despite the above limitations concerning the delivery of web-based GIS, a small number of on-line GIS/MCE applications have been developed. As in the case of general GIS web sites, the greater number of these are descriptive and fall within the dissemination and information categories. These include abstracts, papers, reports, software descriptions and course outlines. A much smaller set of interactive and participatory GIS/MCE web sites have recently come on-line. These include current research at the Joint Research Centre (JRC) at Ispra, Italy and the University of Leeds.

Dissemination and Information-based Web Sites

An extensive search of the web with standard search engines such as AltaVista or Yahoo! reveals a relatively large number of web sites that refer directly or indirectly to GIS/MCE within their text. These fall clearly within the 'dissemination' and 'information' categories described in Table 3.1 and include conference web pages containing papers or abstracts on the topic of GIS/MCE, course outlines with lectures or other materials presented on GIS/MCE, student papers and reports using GIS/MCE approaches, on-line GIS/MCE software manuals and advertising material, and personal web pages of people whose stated interests include GIS/MCE. Some examples of these web sites are given in Table 3.2.

Table 3.2 Example dissemination and information GIS/MCE web pages

Type	Description and URL
Conference	Roosaare (1995) http://www.ciesin.ee/GIS95/abstracts/roos2.html Heywood et al. (1996) http://wwwsgi.ursus.maine.edu/gisweb/spatdb/egis/eg94072.html
On-line paper	Fedra (1995) Decision support for natural resources management http://www.ess.co.at/docs/abs/f95b.html
Course materials	Rossiter (1995) Land Evaluation lecture notes http://wwwscas.cit.cornell.edu/landeal/les_notes/s494ch5p.htm University of Leicester – Geographical Decision-making http://indigo.stile.le.ac.uk/~gis-grp/g705.html
Project reports	Gascoigne (1995) Archaeography of Yorkshire Dales (student report) http://www.geog.leeds.ac.uk/staff/m.blake/magis94/gisjg/exarchy/exarchy.htm Collaborative Spatial Decision Support: Moore et al. (1996) http://weber.u.washington.edu/~tjmoore/csdm.html
Software	IDRISI - GIS software http://www.idrisi.clarke.edu/ AR/GIS - Active Response GIS by CIESIN http://www.ciesin.colostate.edu/argis/
Personal pages	University of Leeds – author's home pages http://www.geog.leeds.ac.uk/staff/s.carver/s.carver.html

Interactive Demonstrations

A smaller number of interactive GIS/MCE sites have recently appeared demonstrating that it is possible to run GIS/MCE software over the web using real data. These tend to concentrate on either demonstrating software or application developments. Two web-based GIS/MCE demonstrations are reviewed here: an application of the reference point method and a new approach to uncertainty in MCE, both of which have been developed at the JRC.

Multi-criteria Decision Aids for Regional Planning

In the context of its research and development work on decision support systems for environmental management the Institute for Systems, Informatics and Safety (ISIS) at the JRC has prepared a web-accessible application linking the techniques of GIS and multi-criteria decision aids (MCDA) for site planning. Through the web interface it is possible to access several layers of geo-referenced information that are relevant for siting problems within a regional planning context. These include population density, distance from protected areas, distance from main roads, altitude, and slope. These layers have been prepared for a particular study area covering the Lombardy region in northern Italy. The aim of this case study is to demonstrate the use and application of the software as a decision support tool addressing the problem of finding new sites for development. The *raison d'être* behind the web site is...

Not to develop an exhaustive study of a real case of site localisation of a specific facility (such as a landfill, an incinerator, a nuclear waste disposal site, a water treatment plant, etc.), but to show how the developed module... can be applied to help in analysis and treatment of site searching problems when environmental issues are concerned. (Menegolo, 1996)

After viewing map images of the GIS data layers the user can use the MCDA software to define suitability surfaces for a particular facility or development. The MCE approach adopted is the reference point method (Wierzbicki, 1980). Within this approach the decision-maker is asked to specify aspiration levels, or reference points, for each factor. The software does this by providing an interface via which a reference point is specified for each of the factors relevant to the siting problem. On choosing 'evaluate' from the menu options the reference point method is used to calculate the multi-criteria distance between each possible site in the geographical region concerned and the reference point (Menegolo and Peckham, 1996). The result is a map of the region in which each pixel is color-coded according to its multi-criteria distance from the reference point, thus enabling the user to identify areas most suitable for the site or development according to the chosen preferences. Application interface and example map are shown in Figure 3.1.

This application is clearly restricted to a particular geographical region and limited set of criteria. However, it does illustrate how GIS and multi-criteria analysis can be rendered accessible and usable over the web via an easy-to-use standard browser interface.

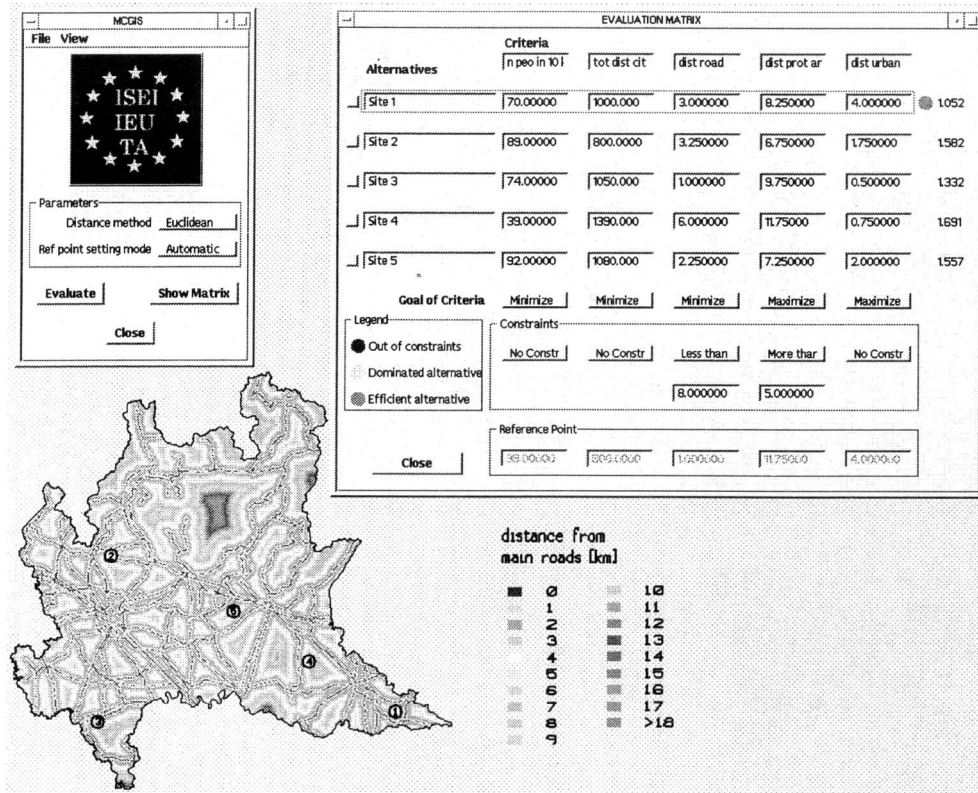


Figure 3.1 GIS/MCDA web interface utilizing the reference point method (after Menegolo, 1996)

Handling Uncertainty in MCE Applications

Another development from the ISIS/JRC is a new multi-criteria method in handling uncertainty, for which a web-based tutorial has been developed. The method is called NAIADE - Novel Approach to Imprecise Assessment and Decision Environments (Munda, 1995; Environmental Information Systems, 1997). The novelty of this method is that it permits the use of information affected by different types and degrees of uncertainty. To do this it can handle the representation of the decision criteria as crisp, stochastic, or fuzzy numbers as well as by ordinal classes expressed as linguistic values (i.e., very good, good, moderate, etc). It also has provisions for evaluation and analysis of the conflicts among different interest groups and possible coalition formation among them.

In the web-based tutorial the user can apply and test the method through an example case study in environmental management based around the 'Boscone Della Mesola' (Mesola forest) in the Po delta area of north east Italy. This is an area of high conservation value (forest ecosystems) and environmental potential (agriculture and forestry). This problem presents the typical ingredients of most environmental policy issues such as several dimensions of assessment, different actors and interest groups with conflicting interests (i.e. conservation, cultivation of crops, grazing and forestry), and of course, uncertainty, which together require multi-criteria analysis in a fuzzy environment. The tutorial explains the problem and the possible courses of action, in the form of five different choice alternatives (business as usual, optimized agriculture, flooding of the Falce Valley, partial flooding in combination with business as usual, partial flooding in combination with optimized agriculture). The evaluation criteria are set out, together with specification of whether they are handled as crisp or fuzzy numbers or by linguistic terms.

Through the web interface the user can vary the values for the criteria, apply the method, and arrive at the ranking of alternatives. It is also possible to specify the preferences of different interest groups for each alternative scenario and thereby arrive at an analysis of the similarities between groups and the possible coalitions that may be formed on the basis of these preferences.

The basis for this, as for any other MCE method, is the evaluation matrix. This is constructed and evaluated using Java applets making the whole system platform independent. Although the example data given are geographical in nature, they are not dependent on any particular GIS platform, thereby making this method easily portable between systems.

Participatory Decision Support Systems

One example of a participatory GIS/MCE system is the Open Spatial Decision-making (OSDM) web-site developed by the author and colleagues at the University of Leeds (Carver et al., 1996; 1997). This is a prototype information system coupled to a GIS/MCE-based spatial decision support system focusing on the controversial problem of where to dispose of nuclear waste in Britain. Selected menus and images from this web-site are shown in Figure 3.2. The system uses spatial datasets and spatial analytical tools to help users explore the decision problem and develop informed personal siting decisions. The aims behind the development of the system are to:

1. experiment with web design for participatory GIS/MCE-based SDSS;
2. inform the public about a particular spatial decision problem; and
3. study the effects of locality, scale, and social context on decisions.

The main experimental objective of the OSDM system is to investigate the potential of web-based GIS/MCE decision support tools for improving public understanding of and participation in important environmental decisions. It is recognized that future systems will need to cope with a range of spatial scales from local to global and need to adopt multi-level adaptive interfaces to allow easy access to information and tools to a wide spectrum of people from different backgrounds. The system needs to be able to cope with a high level of spatiality in the data contained and therefore requires some form of GIS-engine.

One of the key aims of the OSDM system is to provide the public with access to appropriate environmental datasets and analysis tools to enable informed personal decisions to be made about a difficult spatial problem. The system provides direct access to relevant GIS datasets and the analysis tools with which to use them. This functionality is provided by a simple GIS engine and multi-criteria decision tools. The browser interface and web-page design provide an iterative and semi-structured decision path by which to explore the decision problem and experiment with the potential solutions. Key to this is the ability to visualize and understand the relevant spatial and non-spatial factors through the use of a hypertext information system and map images linked to associated metadata. Exploration of the decision space leads to experimentation with the data using the tools provided and ultimately to one or more potential decision solutions.

Certain social and spatial science objectives also underpin the OSDM system. These are to gain specific insights into how users perceive spatial

decision problems and to determine the manner in which the characteristics of the individual may influence the decisions made. This is being done through the analysis of logged decision trees and user profiling. The layout of the web-pages first gets the user to specify their 'best guess' location for a nuclear waste repository based on existing knowledge and perceptions of the problem. This is then compared to their final choice(s) to determine the spatial effect of information, exploration, and experimentation on subsequent decisions. Cross-referencing with user profile information will enable the effects of location, scale, and social context on decision-making behavior to be investigated.

The radioactive waste disposal problem Clearly, radioactive waste disposal is an important and controversial problem, and as such attracts a high level of interest. Radioactive waste disposal is also a very difficult problem; it is political, it is spatial, it involves multiple criteria, and it involves multiple stakeholders. The political risks in developing a new radioactive waste disposal facility are high, as shown by previous failed attempts to find a site for the growing volumes of radioactive waste in Britain (Openshaw et al., 1989). Local people are strongly opposed to any plans for a disposal facility in their area. This phenomenon is often referred to as the NIMBY (Not In My BackYard) syndrome and has led to a series of vociferous and highly politicized anti-dump campaigns.

The geographical problem of finding a suitable disposal site in Britain is accentuated by the small and densely populated nature of the country. International Atomic Energy Agency (IAEA) siting guidelines (1983) state that any disposal facility should be within an area of suitable geology, remote from areas of high population, easily accessible and outside of designated conservation areas. The geography of Britain makes this a particularly difficult spatial problem since there are few suitable geological environments and the areas remote from population are generally remote from access and, more often than not, designated as conservation areas. Thus, the relevant siting criteria are conflicting and therefore require careful evaluation and analysis.

A further complicating factor is the multiple stakeholder nature of the problem. It involves not just the nuclear industry, but also national government, local government, the general public, and various political and environmental interest groups. As a result of the complex political, geographical, multi-criteria and multi-stakeholder nature of the problem, the adoption of a participatory GIS/MCE-based SDSS approach would seem appropriate.

Developing a GIS/MCE approach to siting radioactive waste disposal facilities A prototype GIS-based SDSS was originally developed to address this problem using the PC Arc/Info package and custom MCE routines (Carver, 1991b). This suffered greatly from the fact that it was based on a stand-alone PC, thereby making it difficult to encourage and facilitate widespread use by interest groups and individuals. The OSDM system developed more recently (Carver et al., 1997) applies the same basic principles of GIS and MCE routines, but within a web-based environment. The act of putting the system on-line has the advantage of making it immediately accessible to a global audience.

Table 3.3 Constraint and factor maps used in OSDM

Constraint Maps	Description
Deep geology	Hydrogeological environments considered suitable for deep disposal of low and intermediate level radioactive wastes.
Surface clay geology	Geological environments considered suitable for the near surface disposal of low and short-lived intermediate radioactive wastes.
Population	Areas with population density less than 490 persons per square kilometer; this threshold is derived from UK nuclear power station siting guidelines by Beale (1987).
Conservation	Areas outside of existing conservation areas including national parks, areas of outstanding natural beauty, heritage coasts, national scenic areas, environmentally sensitive areas and regional parks.
Coastal location	Areas within a 10km buffer of the coastline.
Factor Maps	Description
Population density	Population density calculated from Census data.
Population accessibility	Population accessibility function based on distance weighted sum of population totals within 25km radius filter.
Strategic accessibility	Accessibility to waste generating sites based on distance weighted sum of actual and predicted waste arisings.
Rail access	Linear distance from nearest railway line.
Road access	Linear distance from nearest road weighted by road class/capacity.
Conservation area access	Linear distance from nearest conservation area.

The OSDM system is based on the manipulation of raster map images, but without recourse to proprietary GIS software. It is possible to run a GIS like Arc/Info or GRASS across the web as illustrated by example web-sites at ESRI and REGIS. However, it is easier and more practical for the purposes of most web-based GIS/MCE applications to use a custom program to emulate the required GIS and MCE functions. The custom code used in the OSDM radioactive waste disposal example performs two functions; a series of binary map overlays using constraint maps chosen by the user, and a simple MCE routine based on factor maps and weights, again specified by the user. This operates on a series of pre-loaded raster map images in a single operation. The raster maps and their descriptions are listed in Table 3.3.

All the images listed in Table 3.3 are stored at 4 km² resolution. The constraint maps are stored as 0/1 binary raster map images. The factor maps are stored as 0-255 normalized raster map images required by the MCE routine. The MCE routine used is a simple weighted linear summation model, but could equally be substituted by a range of other models suitable for spatial problem solving. From a user's perspective the web-site described here performs several tasks. These are as follows:

1. a data viewer (based on hypertext and in-line images) allowing the browsing of raster map images of the constraint and factor maps included in the system and viewing of text describing their source, meaning and relevance (metadata);
2. an *a priori* site selection map (this is a clickable map that allows the identification of an initial suitable site based on a 'best guess' approach);
3. a data selection and weighting menu which allows the selection of constraint and factor maps and the specification of preference weights for the chosen factor maps;
4. an overlay and MCE routine which runs on the raster map images according to user-specified choices to produce a results map;
5. a results map viewer which allows results to be viewed (this uses continuous shading to show worst to best sites with the very best sites being highlighted), this map is also clickable and allows the re-identification of the suitable site based on the map produced; and
6. a user feedback page which allows users to provide information about themselves and comment on the system and the decision problem.

The system is also accompanied by an information system, based on hypertext and in-line images (Figure 3.2), that describes the various aspects

of the radioactive waste problem. It is suggested that a familiarity with at least the basics of any problem is necessary for effective decision making. In the case of on-line decision support systems like OSDM, an accompanying information system is helpful in providing background knowledge to those who need it.

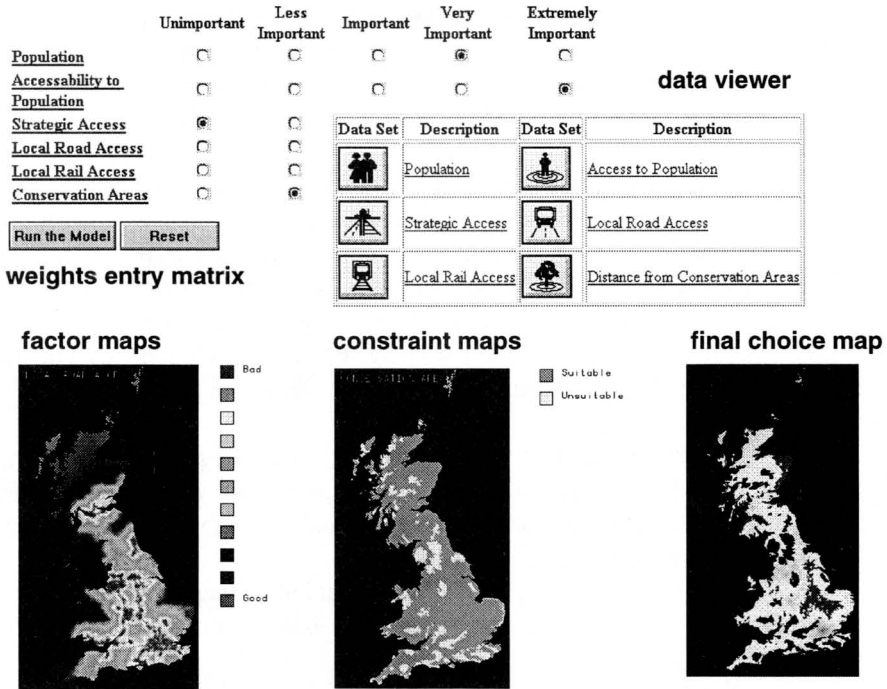


Figure 3.2 Selected menus and images from the OSDM web pages
<http://www.ccg.leeds.ac.uk/mce/mce-home.htm>

All of the above is programmed in HTML v2.0 with the exception of the overlay and MCE routines which are programmed in C. All the menus are intended to be easy to use and full instructions and explanations are provided as appropriate. The decision process outlined by the system is prescriptive (i.e., the data sets are provided and only one model can be used), but is flexible in that it is possible to step backwards through the web pages to any previous stage and review the data or alter selections

made. A key aspect of the system is that it can be populated with new data. All choices and decisions made when the system is used are logged. This enables the reconstruction of decision trees detailing site choices, constraints/factors selected, and assigned preferences/weights. This is being used to construct composite preference maps from which information on most favored sites can be drawn. This kind of information may be useful in helping identifying best compromise and minimum conflict solutions.

GIS/MCE Systems for Participatory Decision Support

It is suggested that the web has an important role to play in developing new approaches to public participation in planning and decision making. In some ways it is possible to see how the various characteristics of the public's role in decision making processes 'map' onto the web, in others the web is such a radical departure from normal procedure that new modes of involvement need to be researched and developed. Many useful examples of how the web can be used to augment current planning and decision making practice already exist. These include instances of planning authorities using the web as a mechanism for communicating strategy documents, informing the public about planning applications, and soliciting feedback about specific issues. These are largely one-way information feeds from the authorities to the public. However, the real potential in the web lies in its ability to handle two-way information flows. Given the necessary interest and foresight the web could be used to foster a closer co-operation between decision makers and the general public. The main area in which web-based approaches to decision making are likely to be most beneficial is in the development of participatory systems designed to improve public involvement in decision making.

In designing web-based GIS/MCE systems for participatory decision support it is possible to draw on a long and varied history of research into decision making and decision support systems. Given the emphasis here on spatial problems, the more recent work on spatial decision support is perhaps most valid. Adopting the general principles described for stand-alone SDSS by Armstrong and Densham (1990), web-based GIS/MCE decision support systems should:

1. be problem specific;
2. combine analytical models and data;

3. allow full exploration of the decision space; and
4. be free from the need for prior knowledge of the system and procedures being used.

In addition, web-based GIS/MCE systems should employ user-friendly interfaces based around common browser formats and should be focused on particular problems that target an enthused and finite audience. If the system is employed in a participatory and consensus-building capacity, then two further aims of such systems are that they should:

1. foster collaboration and participation between groups and individuals with an interest in the decision problem; and
2. generate consensus and/or compromise decisions through shared responsibility in the decision-making process.

Basics of Participatory Systems

Adapting the above principles, the basic stages of web-based collaborative and participatory SDSS can be broken down into three steps or decision stages. These are:

1. exploration of the decision problem;
2. experimentation with choice alternatives; and
3. formulation of one or more decision choices.

These do not explicitly map onto existing decision making processes, rather they are intended to supplement and enhance the traditional role of the public as objectors and/or supporters of planning decisions made on their behalf.

Exploration Exploration of the decision problem is an essential part of the learning process. Having direct and easy access to the information relating to a decision problem is a key element in learning about its various facets. In this context, information should be available on both the spatial and non-spatial aspects of the decision problem. The information is likely to take the form of maps and other spatial data (aerial photographs, satellite images, etc.) as well as multi-media information such as text, sound, images, and video. Multi-media techniques integrated with GIS databases are already quite well developed for stand-alone SDSS (Shiffer, 1994; 1995). The information used should work together to convey the historical and policy context of the decision problem as well its physical, social,

cultural, and economic setting. Existing community and individual ideas and/or perspectives on the decision problem should be presented where known. Through learning about all aspects of a decision problem, it is hypothesized that the user should begin to modify existing ideas and generate new ones. These ideas may be fed back into the information space as and when appropriate.

Experimentation Experimentation with choice alternatives is another essential part of the learning process. Choices of criteria and weighting schema that generate feasible solution sets can be identified by experimentation and fed into the decision space, while choices that generate unfeasible or conflict-laden alternatives can be discarded. This 'What if?' approach is fundamental to many exploratory analyses in GIS. As such, participatory GIS/MCE systems should allow users to:

1. test basic theories/hypotheses regarding specified decision alternatives;
2. develop decision models and/or pathways applicable to the decision problem; and
3. approach consensus and/or compromise through comparison and trade-off with other users' ideas.

Formulation Formulation of choices should aim to maximize consensus and minimize conflict. However, a more realistic approach would be to maximize the acceptability and/or minimize the unacceptability for the individual or interest group. By compiling all valid responses it may be possible to identify the best compromise solution to meet the minimum objectives of all stakeholders. Communication and feedback to all users is essential at this stage (and throughout the decision process) to inform users how and why particular choice alternatives have been identified. Maximizing consensus and/or minimizing conflict throughout the decision process will help maximize the acceptability of final decision and so, it is hoped, minimize any adverse reaction.

Towards 'Virtual Spatial Environments'

Most interactive and participatory GIS/MCE demonstrations/prototypes conform, to a large extent, with the exploration-experimentation-formulation model described above. However, advanced web-based systems designed for real-life participatory and collaborative decision-making applications still need to be developed and tested. Such systems, or

Virtual Spatial Environments (VSEs), are likely to incorporate three essential key elements:

1. virtual spaces for interaction;
2. user-centered adaptive interfaces; and
3. read/write access to relevant information and databases.

These types of systems are 'virtual' in the sense that interaction takes place over the web using many of the tools designed for creating virtual reality (VR) systems. These include virtual chat rooms for group interaction and idea sharing and simulations used to visualize the effects of choices made. In essence, the decisions are real, the people are real, and the data and tools are real, but the methods of exploring and experimenting with data and choices are virtual. The great advantage of this approach is that mistakes can be made in virtual space as part of the 'What if?' exploration and experimentation with the decision problem, before formulating a final choice set for submission into the group decision making process.

Virtual spaces for interaction The virtual spaces for interaction should include both private and open group discussion 'rooms' where users can interact by sharing ideas, exchanging views and contributing information. These are akin to the decision rooms that have long been used in business decision-making, but created on the web in virtual form.

Virtual decision spaces can support both active and passive modes of participation in the form of 'speakers' and 'listeners' and are essential for effective interaction and collaboration between users. One of the advantages of VSEs is that there are few social barriers to participation as tend to exist in public planning meetings which may be dominated by a few forceful individuals with particularly strong views. The availability of virtual spaces may encourage a wider response from the audience by removing some of the barriers to participation present in face-to-face contact and large meetings.

Adaptive interfaces Graphical user interface (GUI) design is fundamental to the development of effective participatory systems. Most web-based interfaces are mouse-driven and based around standard graphical interface devices or 'widgets' such as check-boxes, menus and buttons. Text input from the keyboard is limited. All the main web browsers support this form of interface. Rather, the key problem is user understanding. The user must be able to understand and interact with the interface. This becomes problematic when designing GUIs for general public use due to the wide range of computer experience; from absolute novice to expert. If a GUI is

too simple, it may not offer the flexibility and functionality required to address the decision problem. Alternatively, if the GUI is too complex, novice users may not understand and the learning curve may be too protracted. At the same time it should be recognized that complex decision problems may require complex systems for the decision problem to be addressed fully and without generalizing beyond sensible limits.

The substantive information being presented also needs careful thought. Too many assumptions about the user's knowledge of the decision problem can lead to an incomplete understanding and produce misleading results. Too much basic background information can, on the other hand, seem condescending to more informed users and may cause them to lose interest.

Adaptive user-centered interfaces (AUI) are essential if VSEs are to be accessible to the whole population. It is recognized that the level of education, profession/occupation, age and social background will affect both the level of understanding of a problem and the ability of the user to make best use of the GUI and the system it controls. User interfaces need to take this into account by adapting the complexity and the content of the system to suit the user. Initial user profiling and subsequent user feedback can help indicate the most appropriate level of language and technical complexity incorporated within the interface, information system and decision support elements of the system. Again, care needs to be taken not to allow over simplification of difficult/complex decision problems.

One possible area for research is the development of on-screen 'concept keyboards'. These are based more around physical methods of working and represent a move away from typical menu or command lines modes of human computer interaction (HCI). Here the mode of HCI is based around graphical elements representing real-life workspaces rather than the standard browser interface. With highly spatial decision problems, the most appropriate interface may be the map itself, giving rise to the idea of 'territory as interface'. End users who are not familiar with computers may be much more able to interact with this kind of environment than with standard icons and menus.

Database access Access to relevant information is essential; without it the decision problem cannot adequately be addressed. This may be a problem for certain types of decisions (particularly those involving significant amounts of fuzzy or soft data) or geographical locations where data are scarce. However, data alone are not enough. Spatial and non-spatial data should conform to minimum standards regarding their use and format. These include:

1. use of data model and formats appropriate to the decision problem;
2. use of spatial scales and resolution appropriate to the decision problem;
3. provision of detailed metadata and lineage describing the dataset, its source and method of compilation; and
4. use of appropriate methods of visualization, integration, analysis and interpretation.

Use of intelligent spatial agents may be important in ensuring new users and those unfamiliar with spatial science and GIS do not use inappropriate datasets and/or analyses (Ferrand, 1996). It is recognized that the community of users do themselves represent a potentially vast source of local information, both spatial and aspatial. For this reason VSE architectures should enable the community to further populate the decision space with their own information, giving rise to the concept of the 'community as database'. Clearly, this is important, but does need to be carefully monitored to ensure security is not breached and that the property rights of the contributing parties are safeguarded where necessary.

Improving Public Participation in Spatial Decision Support

In most democratic societies, public involvement in decision-making rises little above the right to object. Web-based SDSS incorporating GIS/MCE methods could act as a focus in moving public involvement in decision making further up the public participation ladder as described by Weidemann and Femers (1993). This upward progression in public involvement in decision-making is shown in Figure 3.3.

Practical and Methodological Barriers to Participation

The democratic ideal of Weidemann and Femers is hampered by a number of practical and methodological problems, not the least of which is the potential for upsetting the current status quo in decision-making power structures. Menegolo (1996) identifies three basic problems facing the development of on-line MCE/GIS approaches to spatial decision-making problems. These include:

1. identifying where the decision-making process actual begins;
2. usability of the model interface; and

3. user understanding and acceptance of the MCE/GIS techniques used.

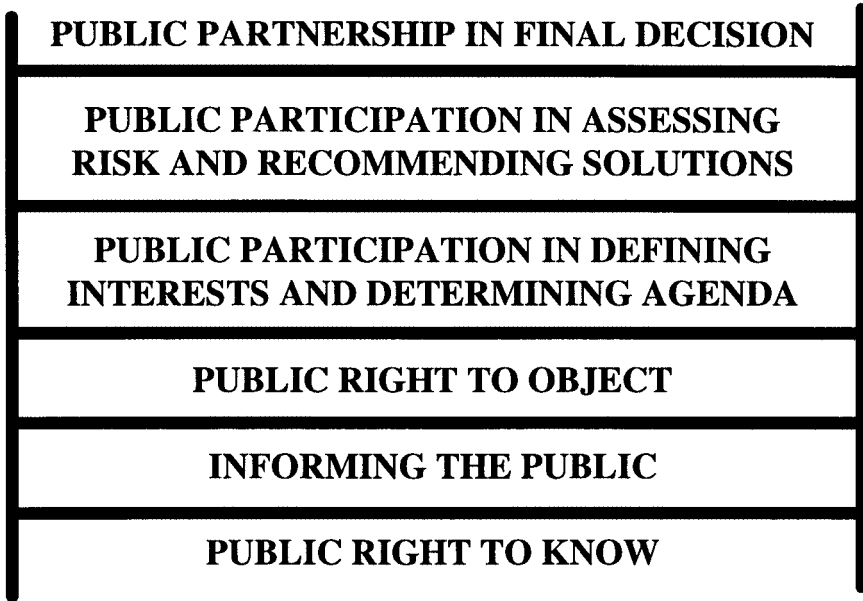


Figure 3.3 The public participation ladder (after Weidemann and Femers, 1993)

The first of these problems is complicated by the lineage of the GIS databases being used. The question often arises (particularly when using existing digital databases) of what decisions have already been made concerning the data before they are made available to the decision maker. For example, which data points have been included, what classification systems used, what method of display has been used, and what decay function selected?

The second refers back to the issue of developing adaptive user-centered interfaces or AUIs. If the interface to the technology is not transparent or it is too complex for the user to understand, the advantage of the system is lost. The third problem is perhaps the most fundamental. If the user does not understand the technology or cannot see how and why the model works, then this can give rise to a crisis of confidence. This problem is succinctly put by Voogd (1983) when referring to the danger of MCE

techniques being used as a 'scientific sauce' to validate and disguise decisions already made, or as Menegolo puts it...

...Spatial analysis is performed through a set of tools that only some GIS magician can effectively use to accomplish this task. The risk is that of moving responsibility down in the decision making hierarchy. Instead of being provided with a support system the decision maker is wary of having lost control of the decision making process. Unfortunately there appears to be no easy solution to this problem. At present great efforts are placed in making GIS easier to use, transparent and flexible. (Menegolo, 1996)

A more immediate and difficult set of problems facing web-based SDSS, however, are the ethical issues concerning under-representation, trivialization of the decision-making process, and bias in system development. These need to be fully researched and systems developed before the web is perceived as a mature enough technology for effective participatory decision support systems to become widespread. These are considered briefly in turn below.

Under-representation is a major problem currently facing the practical, real world use of web-based SDSS. Not everyone at present has access to the web and many people lack essential technical knowledge. This gives rise to the danger of creating an 'Information Underclass' for whom there is no access to information and, as a consequence, lack even a minimum level of understanding of the problem itself (Carver et al., 1997). Apathy and antagonism will surely play an important role here. Many people may simply be uninterested or lack the incentive to participate. On the other hand, many people may be actively hostile to any idea of digital methods of involvement. The potential technological barriers to participation via digital means are, however, likely to reduce with time as more and more people become familiar with computers and their use across a broad spectrum of activities from the home to the workplace. Similarly, the passage of time is likely to see an increased market penetration of the web (or its future equivalent). This is likely to re-shape our use of computers just as television re-shaped our home social lives in the 1960s and 1970s and just as the mobile phone is re-shaping personal communications today. In the short-term, there is likely to be a massive rise in the number of people with connections to the web. Even those who do not own a PC should have easy access to web stations at local libraries, council offices, and other public places, whereas developments in digital TV have the potential to put something akin to the web into many more people's homes. Ensuring sufficient representation via digital means is, however, still a

major problem. While digital methods of participation have the potential to supplement and enhance certain aspects of the decision-making process, it is unlikely that they will entirely replace more traditional face-to-face and paper-based approaches.

Another potential pitfall for web-based GIS/MCE-based SDSS is their potential to trivialize the decision-making process. Decision making is a complex and difficult task, especially at the level of the decisions being discussed here. Public participation in these decisions through web-based systems necessitates some degree of simplification and therefore increases the danger of missing key points or issues and also calls into question the value of the expert knowledge of the trained decision maker. It is perhaps true to say that the best people to make decisions are not the public at large, but the existing decision-making minority who have the required level of expertise and training to do the job effectively. Web-based SDSS may be seen as undermining this authority and replacing it with a 'plug and play' approach not too dissimilar to such products as SimCity. This criticism of 'Nintendo' decision making gives rise to a real worry over misrepresentation of the real views of the population. Users more familiar with computer games than work-place computer applications may not take the problem seriously and may be tempted to play around thereby giving false feedback to the decision makers. Filtering out non-serious responses and validating the seemingly serious is a problem that requires further research.

For many people who are genuinely interested in the decision problem, bias in system development may be a real concern. One of the advantages of the web is its independent nature, but this gives rise to the problem of potential bias in system authoring and control. Taking the radioactive waste disposal problem as an example, a web site authored by the nuclear industry is likely to paint a somewhat different picture of the problem than say one authored by an environmental interest group. The potential for (dis)information in the data, models, SDSS structure, and associated text is enormous. Essentially the onus is on the user to recognize this and place his/her (dis)trust accordingly. This is a basic flaw with any information media, be it the press, television, radio, or the web. Any attempt to police the information provided on the web is against its basic principles of freedom of information and so is either doomed or, if successful, will ultimately kill-off the web.

Conclusions

This chapter has outlined the potential for web-based MCE/GIS in developing participatory approaches to spatial decision support, particularly in addressing specific and important spatial decision problems where a high level of public input is required or deemed appropriate. Selected web-based systems have been described. Comment has been made on research into the potential of web-based GIS/MCE approaches for improving public access to environmental information and involvement in spatial decision-making procedures. Toward this end, this paper has outlined the requirements for web-based participatory decision support systems.

It is the view of the author that despite the potential difficulties such systems have an important role to play in improving public involvement in the decision-making process at a variety of scales from local to global and for a variety of problems with a significant spatial component. The benefits of adopting a web-based approach to GIS/MCE applications and spatial decision making in general focus on the lack of geographical and physical constraints to involvement, the widening of participating public audiences, and the opportunities for public feedback into the decision-making process. The problems focus on two areas: methodological issues of interface design, methods of collaborative working and user understanding; and practical issues of under-representation, trivialization and bias in system development. Although the problems are perhaps not insoluble, much research, technical development, and even political reform are required before such systems become commonplace in real life decision-making situations.

Future developments are likely to see further experimental web-based MCE/GIS approaches to spatial decision support coming on-line in the next few years whilst the wider development of the web as an accepted information medium will see improved public and political awareness of what is possible in this growing field.

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